Feiyang **Chen**

Master of Computer Science · The University of Sydney

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https://freeflyingsheep.github.io | FreeFlyingSheep

Summary

University of Sydney Master of Computer Science graduate with 3 years of Linux kernel and systems experience. Specialised in MIPS/LoongArch enablement, debugging, port validation, and CI/CD, with additional backend and AI application expertise. Independently delivered an LLM news proofreading agent covering fact retrieval, local deployment, asynchronous execution, and observability. Open-source work and technical writing are linked above.

Skills

AI Agent	LangChain, LangGraph, RAG, MCP
Programming	Python, C/C++, React
Data & Middleware	PostgreSQL, Redis
DevOps & Platform	Git, Docker/K8s, Linux, AWS

Work Experience

Suzhou Xinsu Newspaper Network Technology Co., Ltd. *Suzhou, Jiangsu, China*

Software Engineer Intern *Apr. 2026 – May 2026*

- Delivered a news proofreading agent spanning a Vue SDK, fact retrieval, revision generation, and a FastAPI/SQLAlchemy/LangChain backend.
- Deployed Qwen locally and instrumented Langfuse traces for output debugging, task-state visibility, and observability.
- Prototyped drawing/circuit recognition by reproducing research methods and training YOLO, UNet, and HRNet models.

Loongson Technology Corporation Limited *Suzhou, Jiangsu, China*

Kernel Engineer *Dec. 2020 – Aug. 2023*

- Submitted upstream MIPS/LoongArch Linux patches, covering root-cause analysis, fixes, builds, and regression testing.
- Led LoongArch enablement for klibc, Valgrind, and other low-level software; resolved build, instruction, runtime, and performance issues.
- Improved GitHub build, test, and release workflows, increasing delivery reliability and team efficiency.

Jiangsu Lemote Technology Co., Ltd. *Suzhou, Jiangsu, China*

Kernel Engineer Intern *Jun. 2020 – Dec. 2020*

- Developed and debugged Linux kernel code through issue reproduction, log analysis, patching, and validation.
- Ported OpenStack Nova and DevStack to Fedora MIPS, resolving compatibility blockers and validating deployment.

Projects

AI News Proofreading Agent

- Designed an embeddable Vue proofreading SDK for text editing, issue localisation, and revision suggestions.
- Built a FastAPI/SQLAlchemy/LangChain agent for local-corpus retrieval, context construction, and revision generation.
- Added asynchronous execution and error handling; validated traces with local Qwen and Langfuse.

A-Share Stock Analysis Agent Demo

- Built asynchronous ingestion, scheduling, and analysis with FastAPI, SQLAlchemy, and PgQueuer.
- Implemented YAML rules and CNInfo/Yahoo Finance pipelines with retry, rate limiting, and caching.
- Built LangGraph + RAG + SSE analysis with MCP tools; integrated Redis/PostgreSQL, Docker/K8s, and OTel/Prometheus/Grafana.

System Software Adaptation and Tooling

- Added LoongArch64 support to Valgrind; reached initial usability in one month and passed hundreds of regression tests.
- Adapted Linux, klibc, Kcov, and RT-Thread for LoongArch and supported LS2K0300 integration.
- Enabled OpenStack Nova/DevStack on Fedora MIPS and brought up the core service workflow within one month.
- Designed documentation CI/CD with live PR previews and automatic contributor attribution.

Education

The University of Sydney *Sydney, NSW, Australia*

Master of Computer Science *Feb. 2024 - May 2026*

- A. Kumbhakern, **F. Chen**, D. Liyanage, X. Wu, M. A. Alipour, and R. Gopinath, "Dr. DD: 1-Minimal Isolation of Failure Causes via Deferred Restarts," in Proceedings of the 37th IEEE International Symposium on Software Reliability Engineering (ISSRE 2026), 2026. **Accepted / To appear.**

Suzhou University of Technology (formerly Changshu Institute of Technology) *Suzhou, Jiangsu, China*

Bachelor of Computer Science and Technology *Sep. 2016 - Jun. 2020*

- **F. Y. Chen**, W. T. Xu, Z. J. Qian, W. Y. Yang, and W. Liu, "Formal Verification of the Correctness of Operating System on Assembly Layer in Isabelle/HOL," in Artificial Intelligence and Security (ICAIS 2020), Communications in Computer and Information Science, vol. 1252, Springer, 2020, pp. 276–288.